



Continuous practices

ware testing as well as for conducting specialised types of testing such as performance, security or functional. Before choosing the type of cloud-based testing, it is important to consider the benefits and challenges of cloud-based testing.

A cloud-based test grid makes it easy to test an application against a large variety of environments without setting up any infrastructure. A test cloud can collect and display data about test results that can help quality analysis teams to understand why some tests fail for reasons that are hard to track down. It can also help collect and interpret this data from a local test grid.

Cloud-based testing offers a number of advantages for organisations to run tests on a recurring basis as part of continuous delivery pipelines. It allows organisations to significantly reduce the time and cost for software testing.

Benefits of Testing as a Service

TaaS enables running as many tests in parallel as needed. This helps perform

cloud-based tests faster than those that are hosted on local hardware. So, one does not have to wait for one test to complete to use the infrastructure to run the next one. Capacity of the test infrastructure is virtually unlimited in cloud testing, which further increases speed.

Advanced test clouds enable the ability to record videos of tests in action to use at a later time. This can be tremendously useful when there is a need to troubleshoot a test or gain deeper insight.

When a new version of a testing framework is deployed, it needs to add tests for a new type of device or operating system. With a cloud test grid, updates happen automatically, and cloud test vendors handle software updates for the users. This means users can spend more time improving their apps, while someone else keeps their test grid up-to-date. Cloud testing enables them to perform tests on real devices, while also providing access to simulators.

Cloud computing allows testers to increase or decrease computing

resources according to their needs. It lets them use the testing environment they need at the moment and pay only for that environment. Users can have all the software and hardware they might need at their disposal, while only paying for it when they actually use it.

Cloud-based testing avoids any errors made during environment configuration and their repetition across all devices. Software testing in the cloud is available to testers at any-time, anywhere. Cloud-based testing allows software companies to better implement DevOps, which requires collaboration between developers and testers.

Types of cloud testing

The following order may be helpful for testers in getting ready to use cloud-based testing tools:

1. Set clear objectives.
2. Create testing strategy.
3. Plan the infrastructure.
4. Select a reliable provider.
5. Determine service access.
6. Use free trials.
7. Monitor and analyse test results.

In cloud environment, any application can be subjected to three types of testing: functional, non-functional and ability.

Functional testing. This ensures that the software meets functional requirements, which can be of three types:

- System testing that analyses the system's behaviour and design, and how it meets the customer's expectations.
- Acceptance testing that ensures whether the application meets certain needs of its users.
- Integration testing that proves that the application is compatible with different platforms and works well while moving from one cloud infrastructure to another.

Non-functional testing. This allows checking the non-functional aspects of software such as its performance, usability and reliability. This can be done at three levels:

- Business requirement testing that verifies how precisely an application meets the specified business requirements.
- Security testing to ensure that